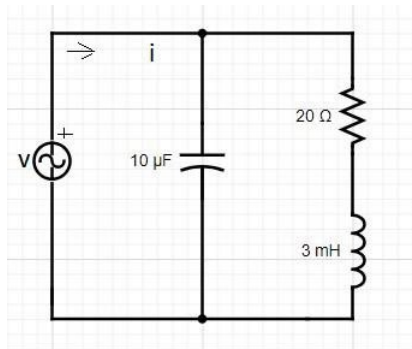
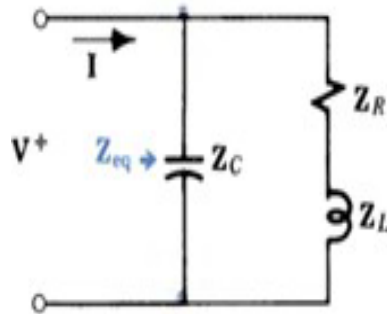


Tutorial - 5

1. A voltage $v = 12\sqrt{2} \cos 5000t \text{ V}$ is applied to the circuit. Find the individual and combined impedances and the current $i(t)$.

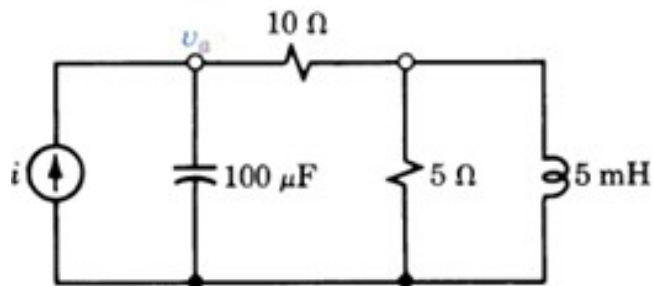


(circuit elements)

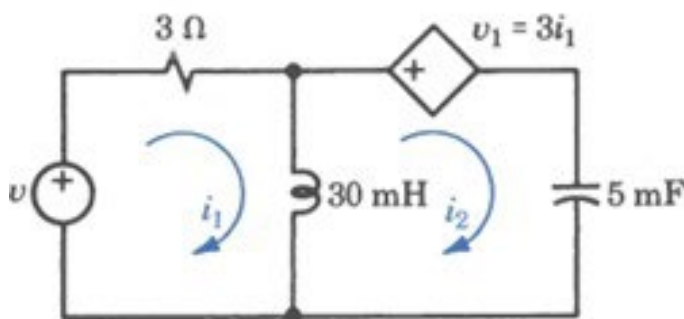


(impedances)

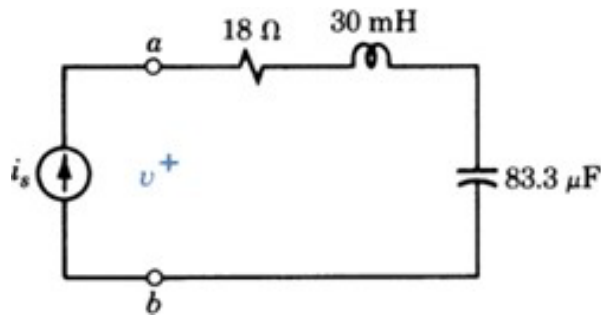
2. Find the steady state node voltage v_a for the given circuit when $i = 10 \cos 1000t \text{ A}$.



3. Find the steady state sinusoidal current i_1 for the circuit when $v = 10\sqrt{2} \cos(\omega t + 45^\circ) \text{ V}$ and $\omega = 100 \text{ rad/sec}$.



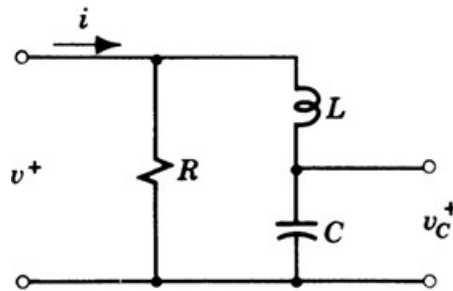
4. Consider the circuit below where $i_s = 120\sqrt{2} \cos(1000t + 90^\circ) \text{ A}$.



Find $v(t)$ where v is the voltage v_{ab} .

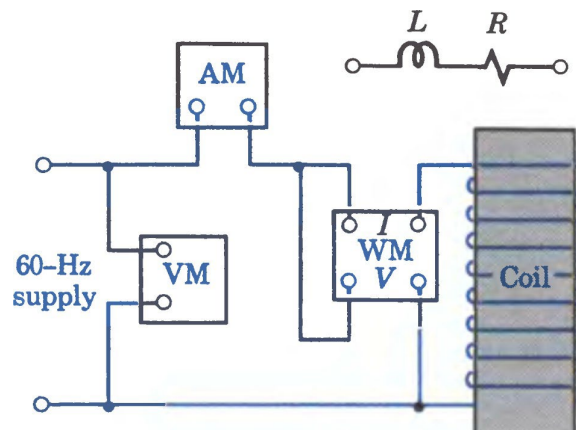
5. In the given circuit below $v = 5\sqrt{2} \cos 5t \text{ V}$, $R = 2 \text{ ohm}$, $L = 1 \text{ H}$ and $C = 0.06 \text{ F}$

- Find the input impedance $Z(j\omega)$
- Determine i , i_R , and i_L and show them with v and v_C on phasor diagram.



- An industrial load consists of the following: 10 kW of lighting at 10 V, a resistance heating unit that draws 150 A at 200 V, and an induction furnace that requires 360 kW and 400 kVAR at 2000 V. Power is supplied to the plant at 40 kV and 60 Hz. Express the total complex power in rectangular and polar form.
- A load connected across a 200-V, 60-Hz line draws 10 kW at a leading pf of 0.55.
 - Determine the current and the reactive power.
 - What series combination of circuit elements is equivalent to this load?

8. A coil is to be represented by a linear model consisting of inductance L in series with resistance R . In the laboratory, when a 60-Hz current of 2 A (rms) is supplied to the coil (Fig. 2), the voltmeter across the coil reads 26 V (rms). A wattmeter indicates 20 W delivered to the coil. Determine L and R .



9. An industrial load consists of 10 kVA of induction motors operating at 0.7 pf lagging and 20 kW of resistance heating. Calculate the total kVA and the plant power factor.
10. Specify the reactive kVA of capacitors required to correct the power factor of the plant in the Problem 9 to 0.85 lagging.